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Data science and machine learning
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Loan Default Prediction using Machine Learning

1. Introduction

Alpha Dreamers Banking Consortium is a financial institution that provides loans and banking services to individuals and businesses. The institution is currently facing a challenge of high loan default rates, which affects its profitability and financial stability.
This project aims to analyze loan application data, identify patterns that influence loan approval, and build a machine learning model to predict whether a loan applicant is likely to default.

2. Problem Statement

The main problem faced by the bank is the high number of loan defaulters. This creates financial risk and reduces overall profit. The goal is to analyze historical loan data and develop a predictive model to help the bank make better lending decisions.

3. Dataset Description

The dataset used in this project contains information about loan applicants such as:
Gender
Marital status
Income
Loan amount
Credit history
Education level
Property area
Loan status (Approved or Not Approved)
This dataset was selected because it closely matches the real-world banking problem of predicting loan default risk.

4. Data Cleaning Process

The dataset was cleaned before analysis and modeling:
Missing values were filled using mode (for categorical data) and mean (for numerical data).
Categorical variables were converted into numerical format using Label Encoding.
Irrelevant columns such as Loan_ID were removed.
This ensured that the dataset was suitable for machine learning algorithms.

5. Exploratory Data Analysis (EDA)

Data visualization was performed using matplotlib and seaborn. The following insights were observed:
Applicants with higher income are more likely to get loan approval.
Credit history is the strongest factor influencing loan approval.
Higher loan amounts increase the risk of default.
Property area and education level also influence loan outcomes.

The dataset shows clear patterns between financial behavior and loan status.
A total of 10 graphs were created to support these insights.

6. Machine Learning Model

A Logistic Regression model was used for this project because it is suitable for binary classification problems such as predicting loan approval (Yes/No).
The dataset was split into training and testing sets to evaluate model performance.

7. Model Evaluation

The model was evaluated using accuracy score and classification metrics.
The results show that the model can reasonably predict loan outcomes based on applicant data.
Evaluation methods used:
Accuracy Score
Confusion Matrix
Classification Report

8. Results and Findings

Credit history is the most important feature in predicting loan approval.
Income and loan amount significantly affect loan decisions.
The model can help reduce risky loan approvals.

9. Conclusion

This project demonstrates how machine learning can be used in the banking sector to reduce loan default risk. By analyzing historical data, the bank can make more informed decisions and improve financial stability.
Logistic Regression proved to be an effective model for predicting loan outcomes.

10. Tools Used

Python
Pandas
Matplotlib
Seaborn
Scikit-learn
Google Colab